Module 0: Magnetism Fundamentals

Physics Quiz - Grade 12 (Review Module)

Instructions

Choose the best answer for each question. This quiz covers the properties of permanent magnets, magnetic field interactions, and the "Big Three" current geometries (Wire, Coil, Solenoid).

Questions

1. If you take a standard bar magnet and cut it perfectly in half between the North and South poles, what is the result?

(A) One piece becomes a pure North monopole, and the other a pure South monopole.
(B) You obtain two smaller magnets, each possessing both a North and a South pole.
(C) The magnetic field disappears completely because the dipole is broken.
(D) The pieces lose their polarity and become non-magnetic iron.

2. Which of the following correctly describes the direction of magnetic field lines **outside** of a permanent magnet?

(A) They originate from the South pole and enter the North pole.
(B) They originate from the North pole and enter the South pole.
(C) They radiate outwards from both poles to infinity.
(D) They circle around the magnet perpendicular to the poles.

3. In a diagram, you see magnetic field lines that are drawn very close together at Point A and far apart at Point B. What does this indicate?

(A) The magnetic field is stronger at Point A than at Point B.
(B) The magnetic field is weaker at Point A than at Point B.
(C) The direction of the field is reversed at Point A.
(D) There is no relationship between line spacing and field strength.

4. According to Oersted's discovery, what is the fundamental source that creates the magnetic field around a wire?

(A) Static electric charges accumulating on the surface.
(B) The iron content inside the copper wire.
(C) Electric current (moving charges).
(D) The voltage difference between the ends of the wire.

5. You are holding a wire carrying current upwards. Using the Right Hand Grip Rule, which way do the magnetic field lines curl?

(A) They point straight up, parallel to the current.
(B) They curl around the wire in the direction of your fingers.

(C) They point radially outward like spokes on a wheel.

(D) They curl in the opposite direction of your fingers.

6. Consider a long straight wire. If you double the distance (r) from the wire, what happens to the magnetic field strength (B)?

(A) It doubles ($2B$).

(B) It stays the same.

(C) It drops to one-fourth ($B/4$).

(D) It is halved ($B/2$).

7. Which of the following creates a **uniform** magnetic field (constant magnitude and direction) inside its core?

(A) A straight wire.

(B) A flat circular loop.

(C) A solenoid (long coil).

(D) A single bar magnet.

8. What is the correct formula for the magnetic field at the **center** of a flat circular coil with radius R?

(A) $B = \frac{\mu_0 I}{2\pi r}$

(B) $B = \frac{\mu_0 N I}{2R}$

(C) $B = \mu_0 n I$

(D) $B = \mu_0 I R^2$

9. How does a DC Clamp Meter measure current flowing through a wire without physically touching the copper?

(A) It measures the heat radiating from the wire.

(B) It detects the magnetic field circling the wire and calculates current from it.

(C) It pierces the insulation to make electrical contact.

(D) It measures the electric field leaking through the insulation.

10. In the formula for a solenoid $B = \mu_0 n I$, what does the variable 'n' represent?

(A) The total number of turns (N).

(B) The length of the solenoid (L).

(C) The number of turns per unit length (N/L).

(D) The radius of the solenoid turns.